

induction pops up again, this time even more significantly than at any previous time in this book. Simply, if a mechanism is fractal it can deliver large values; therefore the incidence of large deviations is possible, but how possible, how often they should occur, will be hard to know with any precision. This is similar to the water puddle problem: plenty of ice cubes could have generated it. As someone who goes from reality to possible explanatory models, I face a completely different spate of problems from those who do the opposite.

I have just read three “popular science” books that summarize the research in complex systems: Mark Buchanan’s *Ubiquity*, Philip Ball’s *Critical Mass*, and Paul Ormerod’s *Why Most Things Fail*. These three authors present the world of social science as full of power laws, a view with which I most certainly agree. They also claim that there is *universality* of many of these phenomena, that there is a wonderful similarity between various processes in nature and the behavior of social groups, which I agree with. They back their studies with the various theories on networks and show the wonderful correspondence between the so-called critical phenomena in natural science and the self-organization of social groups. They bring together processes that generate avalanches, social contagions, and what they call informational cascades, which I agree with.

Universality is one of the reasons physicists find power laws associated with critical points particularly interesting. There are many situations, both in dynamical systems theory and statistical mechanics, where many of the properties of the dynamics around critical points are independent of the details of the underlying dynamical system. The exponent at the critical point may be the same for many systems in the same group, even though many other aspects of the system are different. I almost agree with this notion of universality. Finally, all three authors encourage us to apply techniques from statistical physics, avoiding econometrics and Gaussian-style nonscalable distributions like the plague, and I couldn’t agree more.

But all three authors, by producing, or promoting precision, fall into the trap of not differentiating between the forward and the backward processes (between the problem and the inverse problem)—to me, the greatest scientific and epistemological sin. They are not alone; nearly everyone who works with data but doesn’t make decisions on the basis of these data tends to be guilty of the same sin, a variation of the narrative fallacy. In the absence of a feedback process you look at models and think that they confirm reality. I believe in the ideas of these three books, but not in the way they are being used—and certainly not with the precision the authors ascribe to them. As a matter of fact, complexity theory should make us *more* suspicious of scientific claims of precise models of reality. It does not make all the swans white; that is